

Empowering Learners to Build Production - Ready AI Agents

Applied Agentic AI for SWEs

Week 0 - Mini Project

Solution

Hello Agent - CSV FAQ Agent

1. Step by Step Tasks for the Learner

You can follow these steps to complete the mini project.

Step 1 Prepare your environment

- Make sure you have Python installed
- Create a project folder on your machine
- Create and activate a virtual environment if you want
- Install the required Python packages

Example

```
pip install streamlit pandas langchain langchain-openai
langchain-experimental openai
```

Step 2 Set up your OpenAI API key

- Get your OpenAI API key from your OpenAI account
- You can either paste it into the Streamlit sidebar when the app runs, or set it as an environment variable
- Make sure you keep the key private

Step 3 Create your Streamlit app

Create a file named `app.py` and build a Streamlit layout that includes

- A sidebar that asks for the OpenAI API key
- A file uploader that accepts one or many CSV files
- A main area that shows
    - A title and short description
    - Preview tables for each loaded CSV
    - A text input area for the user question
    - A section to display the AI answer

Step 4 Load and preview the CSV files

- Use Pandas to read each uploaded CSV into a data frame
- Store all data frames in a list
- Show the first few rows to the user as a preview
- Make sure the data looks correct

Step 5 Build a system prompt

Write a clear system prompt for the model.

Your prompt should tell the model to

- Always inspect the data frames using Pandas before answering
- Never answer from general knowledge
- Prefer certain columns like Answer, Policy, Description, or Details when returning text
- For numeric questions, use operations such as sum or mean
- Answer in simple English, quoting actual values

The goal is to constrain the model so that it behaves like a strict data assistant.

Step 6 Create the data frame agent

- Use LangChain to create a data frame agent that wraps the OpenAI chat model
- Pass the list of data frames to the agent
- Set the model to a low temperature value so answers are stable
- Ensure that the agent has permission to run Pandas operations

Step 7 Connect the question input to the agent

- Take the user question from the Streamlit text input
- Combine it with your system prompt into one final query
- Pass that final query to the agent
- Display the agent response in the app

Step 8 Test with sample questions

Try questions for each domain, for example

- Ecommerce
    - What is the return policy for clothing

    - How long does standard shipping take

- Credit card
    - What is the annual fee for the platinum rewards card
    - What is the cash advance fee

- Hospital
    - What are the visiting hours
    - How can I request my medical records

- Software as a service
    - What is the API rate limit for the free plan
    - What support does an enterprise customer get

Check if the answers match the information in the CSV files.

If something looks wrong, inspect the logs and your system prompt.

2. Deliverables

By the end of this mini project you should have

- A working Streamlit app that runs locally on your machine
- The ability to upload the given CSV files
- A simple and clear interface where you can ask questions and read answers
- A short reflection (for yourself) about what you learned

Suggested reflection questions

- How did the system prompt change the behavior of the model
- What was easy in this mini project
- What felt confusing and needs more practice
- How would you extend this app for a real company

3. Stretch Ideas if You Have Extra Time

If you finish before ninety minutes and want to explore more, you can try these optional add ons

- Add a drop down filter to choose which CSV file to query

- Display the row or rows from the table that the model used for the answer
- Log questions and answers in a simple CSV for later review
- Add a short explanation under the answer that says which file and which row was used

These are not required, but they will deepen your understanding of how the agent thinks and how it interacts with the data.

4. Summary

In this Week 0 mini project you will

- Build your first agent style app that reads CSV files
- Work with real looking business data from four different domains
- Map a simple business requirement to a working technical solution
- Gain confidence with Streamlit, Pandas, LangChain, and OpenAI

Treat this project as a friendly warm up. Do not aim for perfection. Focus on understanding each step and seeing the full flow end to end.

Colab Notebook -
https://drive.google.com/file/d/1vb9Y34MaoyrAd0QMYsUGkKW11WZfeSAI/view?usp=sharing

GitHub - https://github.com/ANI-IN/Hello-Agent
Try out these VS Studio Code steps on your local machine